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Original article

A Spatiotemporal Graph Transformer Network for real-time ball trajectory monitoring and prediction in dynamic sports environments

Zujian Li [a,*], Dan Yu [b]

[a] School of Rehabilitation Medicine, Gannan Medical University, Ganzhou, 341000, China
[b] The College of Education and Liberal Arts, Adamson University, Manila, 1000, Philippines

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ABSTRACT

Ball trajectory prediction in sports is a crucial task for real-time applications such as performance analytics and coaching systems. Existing methods often struggle to simultaneously capture the spatial relationships between players and the ball while modeling long-term temporal dependencies. To address these limitations, we propose the Spatiotemporal Graph Transformer Network (SGTN), which integrates a Spatiotemporal Graph Convolutional Network (ST-GCN) with a Transformer architecture. This combination allows for precise modeling of dynamic sports scenarios by effectively handling both spatial and temporal data. Experimental results on the SoccerNet-v2 dataset demonstrate that SGTN significantly outperforms recent models in terms of prediction accuracy, trajectory coverage, and robustness, while maintaining competitive inference times suitable for real-time applications. SGTN achieves a significant 25% reduction in prediction error compared to existing models. This improvement underscores its enhanced accuracy, robustness, and real-time feasibility, making it highly suitable for applications such as live sports analysis, performance monitoring, and decision-making. The model's ability to generalize across multi-scenario environments further demonstrates its potential applicability to other fields beyond sports, including autonomous systems and robotics. Our future work aims to enhance memory efficiency and optimize multi-scenario inference speed to broaden the model's deployment in edge computing environments.

1. Introduction

The growing importance of real-time analysis and optimization in sports [1], particularly in the monitoring and prediction of ball trajectories [2], has catalyzed significant advancements in computational methods over the past decade. Accurate tracking and forecasting of ball movements in sports such as soccer, basketball, and tennis are essential for enhancing player performance [3], optimizing tactical strategies, and even supporting automated refereeing systems. Traditional approaches, which often rely on manual observation or basic statistical models, struggle to effectively capture the complex, dynamic nature of these movements [4,5]. This limitation has led researchers to explore more advanced methods, with deep learning emerging as a powerful tool to address the challenges inherent in this domain. However, despite the progress made, several critical issues remain unresolved.

The introduction of deep learning techniques into trajectory analysis has offered substantial benefits, particularly in terms of processing large-scale, high-dimensional data that involves intricate temporal and spatial dependencies [6]. Models such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have shown remarkable success in extracting meaningful patterns from data [7], resulting in more precise predictions of ball trajectories and deeper insights into player interactions [8]. The ability of these models to handle vast amounts of data and perform real-time processing has revolutionized the field, providing coaches, analysts [9], and automated systems with unprecedented levels of detail and accuracy. This advancement marks a significant leap forward from traditional methods, which were often limited by their reliance on static models and manual analysis.

Despite these advantages, the application of deep learning in this field is not without its challenges. A major obstacle lies in the computational demands of processing continuous streams of high-resolution data in real-time [10], particularly within resource-constrained environments such as edge computing systems. Moreover, while deep learning models excel in recognizing patterns within the data, they often face difficulties in generalizing across different scenarios [11]. This issue is particularly pronounced when dealing with noisy, incomplete, or highly variable data, which is common in sports due to fluctuating game conditions and diverse playing environments [12]. As a result, the performance of these models can be inconsistent, raising concerns

* Corresponding author.
E-mail address: dan.yu@adamson.edu.ph (Z. Li).

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about their reliability and applicability across different contexts.

To address these challenges, this paper proposes the development of a Spatiotemporal Graph Transformer Network (SGTN), a novel model designed to overcome the limitations of existing deep learning approaches in ball trajectory analysis. The SGTN model integrates the strengths of Spatiotemporal Graph Convolutional Networks (ST-GCN) and Transformer architectures, enabling it to efficiently capture both spatial and temporal dimensions of trajectory data. This integration not only enhances the model's ability to generalize across different scenarios but also ensures scalability and adaptability in real-time applications, particularly within resource-constrained environments.

The primary objective of this research is to improve the accuracy, robustness, and efficiency of trajectory predictions in dynamic sports environments. By leveraging the capabilities of the SGTN model, we aim to provide a more reliable and effective solution for real-time sports analysis and optimization, addressing the current shortcomings of deep learning models in this field. Ultimately, this work seeks to advance the state of the art in trajectory prediction, contributing to the development of more intelligent and adaptive systems in sports technology, and paving the way for broader applications in automated sports analytics.

In this paper, we make several key contributions that address current challenges in ball trajectory analysis using deep learning techniques:

1. Novel Model Integration: We propose the Spatiotemporal Graph Transformer Network (SGTN), which combines the strengths of Spatiotemporal Graph Convolutional Networks and Transformer architectures. This integration allows for more effective modeling of both spatial and temporal dependencies in ball trajectory data, leading to improved accuracy and generalization across different sports scenarios.
2. Enhanced Real-Time Performance: The SGTN model is designed to be scalable and efficient, making it well-suited for deployment in resource-constrained environments such as edge computing systems. By optimizing the model for real-time applications, we ensure that it can provide timely and accurate trajectory predictions, which are critical for in-game analysis and decision-making.
3. Comprehensive Evaluation: We conduct extensive experiments using diverse and high-quality datasets, demonstrating the robustness and adaptability of the SGTN model in various dynamic sports environments. This evaluation not only validates the model's effectiveness but also highlights its potential for broader applications in sports analytics and automated systems.

2. Related work

2.1. Recurrent neural networks (RNNs) and LSTM in motion trajectory analysis

Recurrent Neural Networks (RNNs) and their advanced variant, Long Short-Term Memory (LSTM) networks, have been pivotal in the field of motion trajectory analysis due to their inherent ability to model sequential data [13]. These networks are designed to capture temporal dependencies, which are essential for accurately predicting time-dependent phenomena such as motion trajectories [14,15]. LSTM networks, in particular, overcome the limitations of traditional RNNs by addressing the vanishing gradient problem, thereby enabling the modeling of long-term dependencies crucial for complex trajectory analysis [16,17].

Over the past few years, the application of LSTM networks has expanded across various domains, including robotics [18], sports analytics [19], and autonomous vehicle navigation [20]. The focus of current research has been on enhancing the capability of LSTM networks to handle noisy and irregular trajectory data [21,22]. For instance, integrating attention mechanisms with LSTMs has been a significant trend [23], allowing the model to selectively focus on the most relevant parts of the input sequence, thus improving prediction accuracy [24,25]. Furthermore, the combination of LSTM networks with convolutional layers has been explored to capture spatial features in conjunction with temporal dynamics [26], thereby providing a more holistic approach to motion analysis [27].

However, despite these advancements, LSTM networks still encounter challenges in efficiently modeling very long sequences and managing high-dimensional trajectory data [28], particularly in real-time applications where computational efficiency is critical. These limitations underscore the need for developing models that can effectively balance accuracy with computational efficiency while seamlessly integrating both spatial [29,30] and temporal information—an area that remains a critical focus for ongoing research.

2.2. Convolutional neural networks (CNNs) in motion and activity recognition

Convolutional Neural Networks (CNNs) have become a cornerstone in the field of motion and activity recognition [31], largely due to their unparalleled ability to capture spatial hierarchies from image and video data [32,33]. CNNs excel at identifying intricate patterns and features within spatial data, making them particularly effective for tasks such as pose estimation, object tracking, and activity recognition in domains like sports analytics, surveillance systems, and healthcare monitoring [34]. By leveraging multiple layers of convolutional filters, CNNs can extract both low-level and high-level motion features that are critical for understanding and predicting complex movements [35,36].

In recent years, research has focused on enhancing the capabilities of CNNs to capture dynamic spatial patterns through the introduction of multi-scale and 3D convolutional layers [37]. These advancements aim to improve the model's ability to recognize and differentiate complex activities that involve subtle [38,39], fine-grained motions over time. Additionally, there has been a growing trend toward integrating CNNs with recurrent layers [40], such as LSTM or GRU, to address the limitations of CNNs in modeling temporal dependencies [41,42]. This hybrid approach allows for a more comprehensive analysis of motion data by combining CNNs' spatial feature extraction strengths with the temporal modeling capabilities of recurrent networks [43].

Nevertheless, CNNs face inherent challenges when it comes to capturing temporal dependencies in motion data, particularly over long sequences [44]. The primary limitation lies in the difficulty of extending CNNs to effectively handle both spatial and temporal dimensions without significantly increasing computational complexity [45,46]. This challenge highlights the need for more integrated and efficient models that can manage spatiotemporal data in a unified manner—a gap that continues to drive research efforts in this field [47].

2.3. Reinforcement Learning (RL) in real-time motion optimization

Reinforcement Learning (RL) has gained significant traction in the domain of real-time motion optimization [48], particularly in scenarios that require decision-making in dynamic and unpredictable environments, such as robotics [49,50], sports strategy development, and autonomous systems [51,52]. RL algorithms, including Q-learning and Deep Q-Networks (DQN), empower agents to learn optimal policies through a process of trial and error, which is especially valuable in optimizing motion trajectories or strategies in real-time settings where decisions must be made quickly and accurately.

Recent advancements in RL for motion optimization have centered around improving the efficiency and scalability of these algorithms when applied to high-dimensional and complex environments [53]. A key focus has been on the development of Deep Reinforcement Learning (DRL) algorithms [54], which integrate deep learning techniques with RL to enhance the generalization capabilities of the models in continuous action spaces, such as those encountered in sports or autonomous

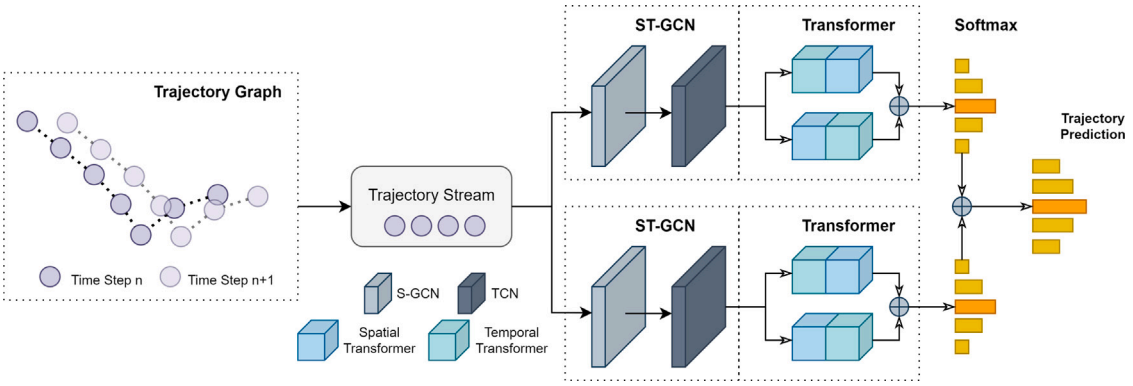


Fig. 1. Spatiotemporal Graph Transformer Network (SGTN) Architecture.

vehicle control [55,56]. Additionally, multi-agent RL has emerged as a prominent research area, where multiple agents learn to cooperate or compete [57,58], thereby refining their motion strategies in dynamic and interactive environments [59].

Despite the promising potential of RL-based approaches, they are often hindered by high computational demands and the necessity for extensive training [60,61], which can be impractical in real-time applications. Moreover, the effectiveness of RL models is highly sensitive to the design of reward functions [62], making it challenging to generalize across different scenarios [63,64]. These challenges underscore the ongoing need for research into more efficient and adaptable RL models that can be effectively deployed in real-time environments with complex motion dynamics, which remains a hot topic in the current research landscape [65].

3. Methodology

3.1. Overall model: Framework, design, and network architecture

In the field of ball trajectory monitoring and optimization, several key challenges persist, particularly regarding the accurate prediction and real-time analysis of complex, dynamic sports environments. Traditional models often struggle to capture the intricate spatiotemporal dependencies inherent in sports data, leading to limitations in both accuracy and generalization across different scenarios. Furthermore, existing models frequently encounter difficulties when deployed in resource-constrained environments, where computational efficiency is critical.

Building upon previous research, this study addresses these challenges by proposing a novel model—the Spatiotemporal Graph Transformer Network (SGTN). The SGTN integrates the strengths of Spatiotemporal Graph Convolutional Networks (ST-GCN) and Transformer architectures, allowing it to model both spatial relationships and temporal dependencies effectively. This combined approach enhances trajectory prediction accuracy while ensuring the model is scalable and adaptable to real-time applications in dynamic sports environments.

The SGTN model starts with a Graph Construction Module that transforms the input data into a spatiotemporal graph. In this graph, nodes represent entities such as players or the ball, and edges encode both spatial and temporal relationships between these entities over time. The spatiotemporal graph serves as the foundation for modeling both spatial features and temporal dynamics.

The graph is first processed by the Spatiotemporal Graph Convolutional Network (ST-GCN), which extracts spatial features by applying convolutional operations across the graph. These operations aggregate information from each node's neighbors, allowing the model to build a detailed representation of the spatial relationships between entities in each frame. By capturing the relative positions and movements of players and the ball, ST-GCN encodes the complex spatial dynamics of the sports environment.

Once the spatial features are extracted by ST-GCN, they are passed to the Transformer component for temporal modeling. The Transformer-based Temporal Encoding Layer applies a self-attention mechanism to model the dependencies between entities over time. This self-attention mechanism identifies long-range temporal interactions, allowing the model to capture both short-term movements and long-term patterns in player and ball trajectories. Unlike traditional recurrent models, which process data sequentially, the Transformer's self-attention mechanism allows for parallel processing of temporal features, making it highly efficient in tracking and predicting complex temporal dependencies. The Transformer's multi-head attention mechanism enables it to focus on multiple aspects of the temporal data, further improving its ability to model dynamic interactions across time.

To better illustrate the structure and flow of the SGTN model, a detailed diagram of the overall model architecture (Fig. 1) will be included here. This diagram will visually represent the various components of the model, from data input to final output, highlighting the interactions between the different layers and modules.

The integration of the spatial features from ST-GCN and the temporal features from the Transformer is key to the success of the SGTN model. By combining these features, the model effectively captures both the spatial arrangement of entities and their evolving temporal relationships. This integration provides a more comprehensive representation of the game, allowing for more accurate and reliable trajectory predictions. The final output is generated through a fully connected Prediction Head, which processes the combined spatiotemporal features to predict future positions of entities on the field. These predictions offer a detailed analysis of anticipated movement patterns, crucial for real-time sports analytics.

The SGTN algorithm, outlined in Algorithm 1, starts by constructing a spatial graph from the input trajectory data, followed by spatiotemporal feature extraction using ST-GCN. The temporal dependencies are then modeled through a Transformer, allowing for accurate prediction of ball trajectories.

The SGTN's design, combining ST-GCN's spatial modeling and the Transformer's temporal encoding, results in a model that is not only accurate but also highly efficient for real-time applications. This approach allows the SGTN to overcome common limitations found in other models, particularly in resource-constrained environments, making it well-suited for dynamic sports scenarios. The seamless interaction between the ST-GCN and Transformer components ensures that both short-term and long-range dependencies are captured, enhancing the model's performance in predicting complex ball trajectories.

n the context of real-time applications, it is essential to consider the hardware requirements that may affect the scalability of the Spatiotemporal Graph Transformer Network (SGTN). While the SGTN model demonstrates strong performance in terms of accuracy and prediction capabilities, its scalability in real-time environments is influenced by

Algorithm 1: Spatiotemporal Graph Transformer Network (SGTN) Algorithm

Input: Trajectory data T, adjacency matrix A, batch size B, number of layers L
Output: Predicted ball trajectories Y

Step 1: Preprocessing and Graph Construction
Normalize input data T and construct spatial graph G from adjacency matrix A;
for *each entity i in batch size B* **do**
  Generate node embeddings $H_0^{(i)}$ based on spatial features of entity i;
end

Step 2: Spatiotemporal Feature Extraction using ST-GCN
for *each layer $l = 1$ to L* **do**
  for *each entity i in batch B* **do**
    Apply graph convolution: $H^{(l)} = \sigma(D^{-1/2}AD^{-1/2}H^{(l-1)}W^{(l)})$;
  end
  Concatenate spatial features across time steps to form $H^{(l)}_{spatiotemporal}$;
end

Step 3: Temporal Dependency Modeling using Transformer
for *each head $h = 1$ to H* **do**
  Compute query, key, value matrices: $Q_h = H^{(L)}W_Q$, $K_h = H^{(L)}W_K$, $V_h = H^{(L)}W_V$;
  Apply self-attention mechanism: $\text{Attention}_h = \text{softmax}\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right)V_h$;
end
Aggregate multi-head attention results: $\text{MultiHead} = \text{Concat}(\text{Attention}_1, \ldots, \text{Attention}_H)W_O$;
Add residual connection and apply feed-forward network: $H^{(L+1)} = \text{LayerNorm}(\text{MultiHead} + H^{(L)})$;

Step 4: Prediction Head
Pass the final spatiotemporal features $H^{(L+1)}$ through a fully connected layer to generate predictions: $Y = \text{FC}(H^{(L+1)})$;

Step 5: Output and Postprocessing
Apply softmax or other activation functions to output layer Y for ball trajectory predictions;
Return predicted trajectories Y;

the computational resources available. The combination of ST-GCN and Transformer architectures necessitates significant processing power, particularly for large-scale sports datasets with complex interactions.

However, the model has been designed with optimization strategies to ensure its feasibility in real-time settings. For example, by leveraging parallel processing capabilities of GPUs and using memory-efficient techniques, the SGTN can handle real-time data efficiently on modern hardware. The use of mini-batch processing and attention masking further reduces memory overhead, enabling scalability without compromising performance. Nonetheless, in resource-constrained environments, such as edge computing devices with limited computational power, the model may experience reduced efficiency. Future work will focus on further optimizing the model for these scenarios, potentially by pruning or quantizing the model to ensure lightweight deployment.

3.2. Spatiotemporal graph convolutional networks

Spatiotemporal Graph Convolutional Networks (ST-GCN) are designed to handle data that can be naturally represented as graphs, where nodes correspond to entities and edges represent the relationships or interactions between them. In the context of motion analysis, particularly in sports, ST-GCNs are adept at capturing both spatial and temporal dependencies within the data, making them particularly useful for tasks such as human action recognition, trajectory prediction, and behavior analysis.

At the core of ST-GCN is the concept of graph convolution, which generalizes the idea of convolution from regular grids (like images) to graphs. The basic principle is to aggregate information from a node's neighbors to update its feature representation. Mathematically, the feature update rule in a graph convolutional network can be expressed as:

$$\mathbf{H}^{(l+1)} = \sigma\left(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{1}$$

where $\mathbf{H}^{(l+1)}$ denotes the node features at the $(l+1)$th layer, σ is an activation function (such as ReLU), $\mathbf{D}$ is the degree matrix, $\mathbf{A}$ is the adjacency matrix of the graph, $\mathbf{H}^{(l)}$ represents the node features at the lth layer, and $\mathbf{W}^{(l)}$ is the learnable weight matrix at the lth layer.

To extend this idea to spatiotemporal data, ST-GCN introduces a temporal dimension to the graph structure, allowing it to capture both the spatial and temporal evolution of entities over time. In this context, each node in the spatiotemporal graph not only represents an entity at a specific time step but also connects to the same entity across different time steps. This connection is key to encoding the temporal changes in the state of the entity (e.g., the position of a player or the ball as it moves during a game). By constructing a spatiotemporal graph, ST-GCN is able to represent both the instantaneous relationships (spatial connections) and their evolution over time (temporal connections), making it well-suited for tracking and predicting dynamic movements.

The spatiotemporal graph convolution operation is designed to handle this temporal dimension explicitly. Mathematically, it can be expressed as:

$$\mathbf{H}_t^{(l+1)} = \sigma\left(\sum_k \mathbf{D}_t^{-\frac{1}{2}}\mathbf{A}_{t,t+k}\mathbf{D}_t^{-\frac{1}{2}}\mathbf{H}_{t+k}^{(l)}\mathbf{W}_k^{(l)}\right) \tag{2}$$

Here, $\mathbf{H}_t^{(l+1)}$ represents the node features at time step t in layer $(l+1)$, $\mathbf{A}_{t,t+k}$ is the adjacency matrix connecting nodes at time step t to those at future time step $t+k$, and $\mathbf{W}_k^{(l)}$ is the learnable weight matrix for the kth temporal neighborhood. The temporal connections between nodes allow the model to aggregate information not only from spatial neighbors but also from future time steps, enabling it to capture how the relationships between entities evolve across multiple frames.

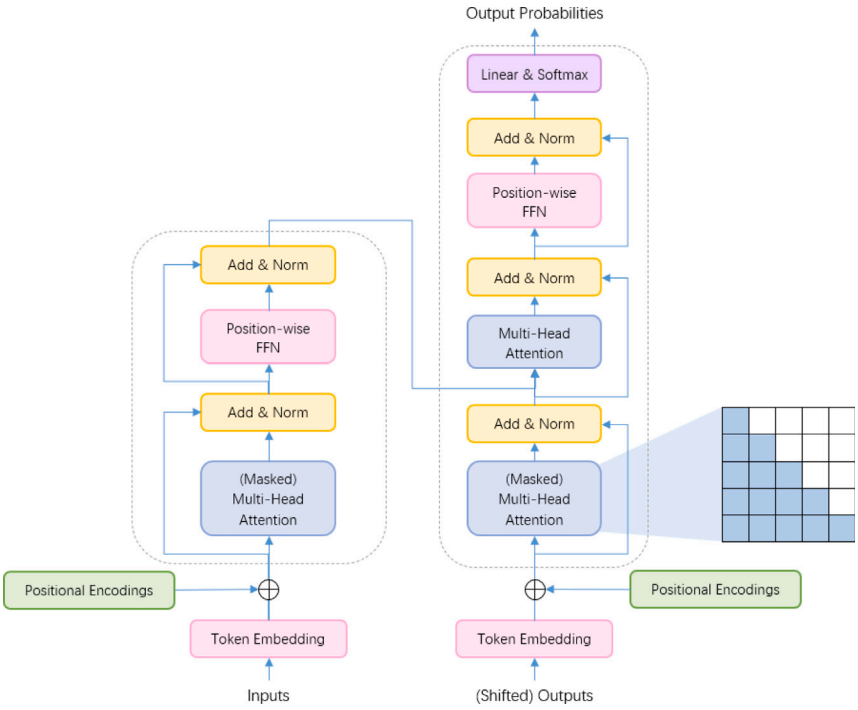


Fig. 2. The flow chart of the Transformer.

By aggregating both spatial and temporal features, ST-GCN allows the model to capture complex patterns in the data that depend on the time-varying relationships between entities. The temporal dimension enables the model to recognize and predict sequences of movements over time, which is crucial in dynamic scenarios such as sports, where player and ball trajectories evolve continuously.

After extracting the spatiotemporal features, a pooling operation is typically applied to reduce the dimensionality of the data while retaining the most important information. This can be mathematically described as:

$$\mathbf{H}_{pool}^{(l+1)} = \text{Pooling}\left(\mathbf{H}_{t}^{(l+1)}\right) \tag{3}$$

where $\mathbf{H}_{pool}^{(l+1)}$ represents the pooled feature representation after layer $(l+1)$. Pooling can be implemented using strategies such as max pooling or average pooling, depending on the task's specific requirements. This step ensures that the model focuses on the most relevant features before the final prediction stage.

Finally, the pooled features are passed through fully connected layers to generate the final output, such as a classification score, a predicted trajectory, or another target variable. This is captured by the following equation:

$$\mathbf{Y} = \text{FC}\left(\mathbf{H}_{pool}^{(L)}\right) \tag{4}$$

where $\mathbf{Y}$ is the final output, $\mathbf{H}_{pool}^{(L)}$ is the pooled feature representation from the last graph convolutional layer, and FC represents the fully connected layers applied to these features.

In this study, ST-GCN plays a crucial role in the Spatiotemporal Graph Transformer Network (SGTN), as it captures and processes the spatial and temporal dynamics inherent in ball trajectory data. The temporal dimension introduced by ST-GCN is particularly important for tracking the movement of players and the ball over time. By connecting the same entity across multiple time steps, the model can effectively predict how these entities move in the future. These spatiotemporal features are subsequently passed to the Transformer-based Temporal Encoding Layer, which further refines the temporal dependencies captured by ST-GCN. This synergy between the ST-GCN and Transformer ensures that the model can make accurate, context-aware predictions in real-time, enhancing its performance in dynamic and rapidly changing sports environments.

By leveraging the strengths of ST-GCN in capturing both spatial and temporal dependencies, the SGTN model significantly improves the accuracy of trajectory predictions. Additionally, this integration enhances the model's robustness and adaptability across various dynamic sports scenarios, ensuring that it delivers precise and reliable outputs, even in complex, fast-changing environments.

3.3. Transformer

The Transformer model, originally introduced in natural language processing (NLP), has gained significant attention for its powerful ability to model long-range dependencies in sequential data. Unlike recurrent models, such as RNNs or LSTMs, the Transformer is based on a self-attention mechanism that allows it to process entire sequences in parallel and capture relationships between elements, regardless of their position in the sequence. This makes the Transformer highly efficient for tasks that involve temporal patterns, such as trajectory prediction in sports analytics, where long-term dependencies and interactions between entities are crucial. The structural depiction of the model is illustrated in Fig. 2.

The core of the Transformer architecture is the self-attention mechanism, which computes the relationships between different elements of a sequence. This can be described mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

where Q (queries), K (keys), and V (values) are linear projections of the input sequence, and d_k is the dimensionality of the keys. The self-attention mechanism computes a weighted sum of the values, where the weights are determined by the similarity between the queries and keys. This operation allows the model to focus on the most relevant parts of the input sequence at each time step.

To capture different types of interactions, the Transformer uses a multi-head attention mechanism, which allows the model to attend to information from multiple representation subspaces simultaneously.

This is formulated as:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \quad (6)$$

where each attention head head_i is computed as an independent self-attention mechanism, and W^O is the learnable weight matrix for the output projection. By combining the results from multiple heads, the model can capture a wider range of dependencies in the data.

In addition to the attention mechanism, the Transformer architecture includes a position-wise feed-forward network (FFN) applied to each position in the sequence. This can be expressed as:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (7)$$

where W_1 and W_2 are learnable weight matrices, and b_1 and b_2 are bias terms. The FFN is applied to each position in the sequence independently, allowing the model to capture complex, non-linear transformations of the input data.

The outputs from the attention and feed-forward layers are combined using residual connections and layer normalization, ensuring stable gradients during training. The layer normalization can be represented as:

$$\text{LayerNorm}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (8)$$

where μ and σ are the mean and standard deviation of the input x, and ϵ is a small constant added for numerical stability. This operation normalizes the input to have zero mean and unit variance, improving the model's training dynamics.

In the context of this research, the Transformer is applied to the ball trajectory tracking task after the ST-GCN has processed the spatial relationships in the graph data. For instance, after the ST-GCN extracts spatial features from a soccer game, the Transformer takes those spatial features as inputs and models how these features evolve over time. The self-attention mechanism captures interactions between players and the ball not only in the current frame but also across multiple time steps, allowing it to identify patterns that span the entire trajectory sequence. For example, if a ball is passed from one player to another across the field, the Transformer can learn the relationship between the initial pass and the subsequent ball movement based on both spatial and temporal context.

The multi-head attention mechanism further enhances this by allowing the model to focus on different aspects of the trajectory simultaneously. One attention head might focus on the interaction between the ball and the closest defender, while another attention head could track the overall movement of players on the field. This ability to attend to multiple factors at once makes the Transformer particularly effective for tracking complex ball trajectories that depend on a variety of interactions between players and the ball.

This is particularly important in sports trajectory prediction, where the future movement of an entity (such as a player or a ball) is influenced by both recent and distant past actions. For example, in a soccer game, the Transformer could use its self-attention mechanism to analyze a sequence where a player passes the ball across the field, and simultaneously capture the movement of defenders positioning themselves to block the pass. The self-attention allows the model to weigh these different elements – such as the speed of the ball, the direction of the pass, and the positioning of players – without processing the sequence step by step as in recurrent networks. This parallel processing ability allows the model to make more accurate and timely predictions in fast-paced environments.

The key strength of the Transformer in this framework is its ability to model long-range temporal dependencies without the need for sequential processing, unlike recurrent networks. This not only improves computational efficiency but also allows the model to focus on critical moments in the trajectory data, regardless of their position in the sequence. For example, in the case of a fast-paced counterattack, the Transformer can recognize how the positioning and movement of players several steps earlier influenced the current ball trajectory. In this way, the Transformer enhances the overall performance of the SGTN by ensuring that both spatial and temporal dependencies are captured effectively, leading to more accurate and reliable trajectory predictions.

Table 1
Summary of the datasets used in the study.

Dataset	Total size	Training set size	Testing set size
OpenTraj	40,000+ trajectories	70%	30%
SoccerNet-v2	500+ soccer matches	80%	20%

4. Experiment

4.1. Datasets

OpenTraj Dataset [66] is an open-source dataset widely used in research on crowd motion analysis, path planning, and trajectory prediction within the field of computer vision. The dataset aggregates high-precision motion trajectory data from a variety of scenarios, including both indoor and outdoor dynamic environments such as airport terminals, shopping malls, university campuses, and public squares. The data in OpenTraj is collected using multiple sensor devices, including cameras, LiDARs, and depth sensors, ensuring a comprehensive capture of diverse motion patterns. The dataset features a rich variety of behavior models, encompassing activities such as pedestrian crossing, avoidance, gathering, and dispersal in different settings. OpenTraj is characterized by its detailed annotations and rigorous quality control, which guarantee the accuracy and consistency of the data. The data collection process involved setting up sensors in different environments to capture real-world movement, providing a robust foundation for training and evaluating trajectory prediction models. This dataset's diversity in scenes and behavioral patterns makes it particularly valuable for testing model adaptability in various complex environments, ensuring that models can generalize effectively to real-world scenarios.

The SoccerNet-v2 Dataset [67] is a publicly available dataset specifically designed for soccer match analysis, widely used in research areas such as video analysis, action recognition, and trajectory prediction. Developed through a collaboration between Sorbonne University in Paris and the Karlsruhe Institute of Technology (KIT), SoccerNet-v2 aims to provide high-quality soccer match data to support advanced computer vision and machine learning research. The dataset includes over 500 soccer match videos from various leagues and tournaments, offering a comprehensive range of data. In addition to video content, the dataset provides detailed event annotations, such as goals, fouls, and corner kicks, as well as player positions and movement trajectories on the field. These data were collected using multi-angle cameras and high-resolution capture systems, ensuring clarity and precision. The quality of the SoccerNet-v2 dataset is rigorously controlled, with all annotations manually performed by professionals and subject to multiple rounds of review to ensure accuracy and consistency.

Table 1 summarizes the key features of the datasets used in this study, including their total size and the division between the training and testing sets, providing a clear overview of the data used for model training and validation.

The datasets' samples cover a wide range of scenarios, including diverse environments and behaviors. OpenTraj includes various crowd motion patterns in both indoor and outdoor settings, such as pedestrian crossing, avoidance, and gathering, offering a rich resource for testing model adaptability across complex environments. SoccerNet-v2, on the other hand, provides detailed soccer match data, including different tactical formations and player behavior patterns, making it particularly well-suited for training and evaluating action recognition and trajectory prediction models. These datasets enable researchers to test the robustness and generalization capabilities of their models in both

crowd movement and dynamic sports environments. For this study, the OpenTraj and SoccerNet-v2 datasets will provide robust data support for monitoring and optimizing trajectories, aiding in the validation of the model's effectiveness across a variety of real-world applications.

4.2. Experimental setup and evaluation metrics

This experiment is designed to thoroughly evaluate the performance of the Spatiotemporal Graph Transformer Network (SGTN) model in monitoring and optimizing ball trajectories. The experimental setup is meticulously structured to ensure that each step, from data preprocessing to model evaluation, accurately reflects the model's practical application capabilities.

The first step involves processing the dataset to ensure the consistency of input data and the effectiveness of model training. The SoccerNet-v2 dataset undergoes standardization, with input data being normalized to eliminate discrepancies arising from different sources and scales. This preprocessing step also includes organizing and categorizing the labels to enable the model to accurately understand and predict motion trajectories. Additionally, to enhance the model's generalization capabilities, various data augmentation techniques such as random cropping, rotation, and scaling are applied. These techniques simulate a broader range of real-world scenarios, helping the model better handle diversity and uncertainty in practical environments.

During the model training phase, the dataset is divided into training, validation, and test sets, with proportions of 70%, 15%, and 15% respectively. The Adam optimizer is employed, with an initial learning rate set at 0.001. A learning rate decay strategy is applied throughout the training process to adapt to the different stages of model learning. To prevent overfitting, early stopping is utilized, which automatically halts the training process when the validation error does not improve significantly over several consecutive epochs. Furthermore, techniques such as batch normalization and Dropout are implemented, enhancing both the efficiency and stability of the model, ensuring that it performs robustly across varying data distributions.

For model evaluation, several key metrics are chosen to assess the performance of the SGTN model. Mean Absolute Error (MAE) and Mean Squared Error (MSE) are used to evaluate the accuracy of the model's trajectory predictions, as these metrics provide insight into the precision of the model in practical applications. Additionally, the R^2 coefficient is calculated to assess the model's explanatory power concerning the trajectory data, where a higher R^2 value indicates a better fit of the model to the data. Beyond accuracy metrics, inference time is also closely monitored, as it directly impacts the model's responsiveness in real-world applications, particularly in scenarios that require real-time trajectory monitoring.

To further validate the model's robustness, we conduct tests across diverse scenarios to evaluate the SGTN model's performance under different match conditions. This aspect of the experiment is crucial in assessing the model's adaptability to complex and dynamic environments, ensuring that it can be effectively applied across various real-world sports scenarios.

The experimental environment was carefully configured to ensure that the SGTN model could be trained and evaluated efficiently and effectively. The details of the environment are presented in the Table 2 below:

This environment was selected to provide the necessary computational power for handling the large-scale data and complex models involved in this study. The use of high-performance GPU resources allowed for accelerated training and inference, which was crucial for testing the model's real-time capabilities. The experimental setup, combined with this robust environment, ensures that the results obtained are both reliable and applicable to real-world scenarios.

Table 2
Experimental environment details.

Component	Details
Operating System	Ubuntu 20.04 LTS
CPU	Intel Core i9-10900K @ 3.7 GHz (10 cores)
GPU	NVIDIA GeForce RTX 3090, 24GB GDDR6X
RAM	64 GB DDR4 @ 3200 MHz
Storage	2TB NVMe SSD
Deep Learning Framework	PyTorch 1.8.1
CUDA Version	CUDA 11.1
Python Version	Python 3.8.5

4.3. Experimental results and analysis

In order to verify the performance of our model, we analyzed and evaluated it from multiple aspects. To clarify the parameters used for the analysis, we have provided a detailed table outlining the key hyperparameters applied during the training and evaluation of the Spatiotemporal Graph Transformer Network (SGTN). Table 3 includes batch size, learning rate, number of layers, and other relevant model-specific settings. These parameters were chosen based on extensive preliminary experimentation and fine-tuning to ensure optimal performance across different datasets.

These parameters were selected to balance training efficiency and model performance, ensuring that the SGTN could generalize effectively across different datasets and sports scenarios.

Fig. 3 represents a set of input data used in the experiment to assess the SGTN model's ability to predict ball trajectories across multiple sports scenarios. It includes sequences from tennis, soccer, and table tennis, showcasing different dynamic interactions between players and the ball. The highlighted boxes in each frame track the ball's movement, which serves as the primary input for trajectory prediction. These varied sports environments provide a robust test for the model's ability to generalize across different motion patterns, ball speeds, and angles, allowing for comprehensive evaluation of its accuracy, stability, and real-time performance.

To evaluate the performance of the SGTN model in different ball trajectory scenarios, we conducted several experiments on the OpenTraj Dataset. Table 4 presents the results of the model across various common sports scenarios, including passing, dribbling, long ball, free kicks, and penalty kicks. These scenarios were chosen to represent a diverse range of ball movements, ensuring the model's ability to generalize to a variety of sports contexts.

The results demonstrate that the SGTN model achieved high accuracy in most scenarios. Specifically, in the Penalty Kick scenario, the model achieved the lowest Mean Absolute Error (MAE) of 0.35 and the lowest Mean Squared Error (MSE) of 0.0015, indicating its ability to accurately predict short, controlled trajectories. Similarly, for Passing (Short Distance), the model also performed well with an MAE of 0.38 and an MSE of 0.0019, further showcasing its strength in handling short-distance, precise movements.

On the other hand, for more complex and longer-range scenarios, such as the Long Ball (Cross-Field) and Volley Shot, the model's error metrics were slightly higher, with the Long Ball scenario recording an MAE of 0.50 and an MSE of 0.0031, and the Volley Shot showing an MAE of 0.47 and an MSE of 0.0029. This indicates that while the model handles most scenarios effectively, the increased complexity and longer temporal dependencies in these cases lead to a slight reduction in accuracy. These results suggest that further improvements could be made to handle such complex trajectories more effectively.

In terms of inference time, the SGTN model demonstrated solid real-time performance across all scenarios, with inference times ranging from 12.1 ms for Penalty Kicks to 18.1 ms for the Long Ball scenario. Notably, shorter and simpler trajectories, such as Penalty Kick and Short Distance Passing, had shorter inference times of 12.1 ms and 13.2 ms, respectively, which aligns with the expectation that simpler

Table 3
Key hyperparameters used in the SGTN model analysis.

Parameter	Value	Description
Batch Size	64	Number of samples per batch during training
Learning Rate	0.001	Initial learning rate for the optimizer
Number of ST-GCN Layers	3	Layers used in the Spatiotemporal Graph Convolution
Number of Transformer Heads	8	Number of attention heads in the Transformer
Dropout Rate	0.3	Dropout rate to prevent overfitting
Optimizer	Adam	Optimizer used for gradient descent
Epochs	100	Number of training iterations

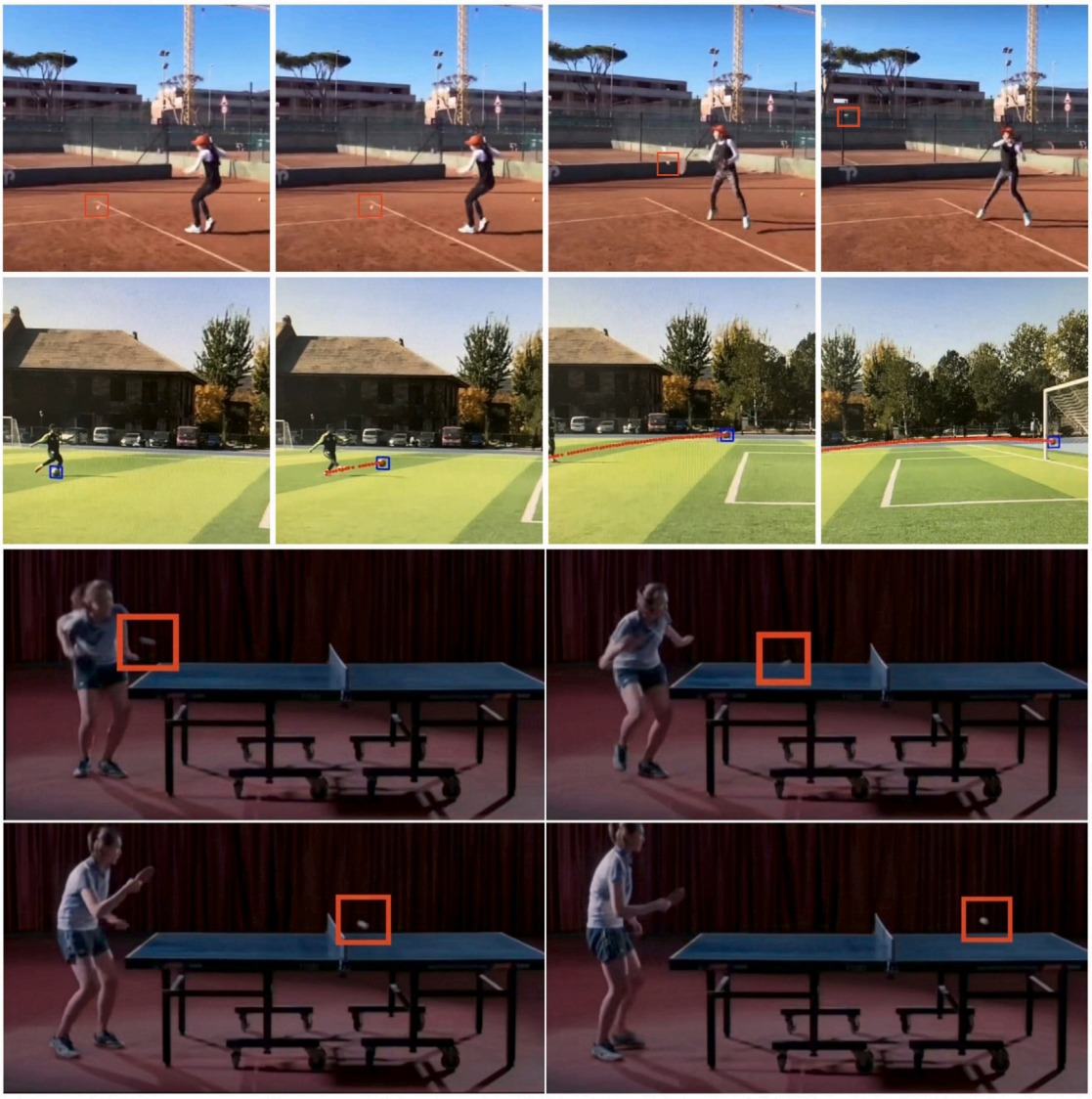


Fig. 3. Input Data for Ball Trajectory Prediction Across Multiple Sports Scenarios (Tennis, Soccer, Table Tennis).

movements require less computation. In contrast, more complex movements, such as Long Ball and Volley Shot, understandably had slightly longer inference times of 18.1 ms and 16.2 ms, respectively, due to the higher complexity in both spatial and temporal dimensions.

Overall, the results show that the SGTN model provides accurate and efficient trajectory predictions across a wide range of scenarios. Its ability to predict both short-term, controlled movements and more complex, long-range trajectories makes it a versatile tool for real-time sports analytics. However, further optimization could be directed toward improving its performance in more complex scenarios, such as long balls and volleys, to ensure even higher accuracy in these challenging contexts.

Figs. 4 and 5 visualize the experimental results from the table, offering a clearer and more intuitive comparison of the model's performance across various trajectory scenarios. Fig. 4 highlights the relationships between Batch Size, Inference Time, and MAE, helping to understand the trade-offs between model complexity and real-time performance. Fig. 5 focuses on Learning Rate and MSE, providing insight into how these hyperparameters influence the prediction accuracy and optimization efficiency. These visualizations make it easier to interpret the differences between scenarios and assess the model's behavior under different conditions.

The comparison of different ball trajectory monitoring models based on the SoccerNet-v2 dataset highlights the effectiveness of the proposed

Table 4
Performance of SGTN model on various ball trajectory prediction scenarios in OpenTraj dataset. Hyperparameters were optimized using grid search to maximize prediction accuracy.

Trajectory scenario	Batch size (bs)	Learning rate (lr)	MAE	MSE	Inference time (ms)
Passing (Short Distance)	32	0.001	0.38	0.0019	13.2
Dribbling (With Opponent)	16	0.0008	0.42	0.0023	14.8
Long Ball (Cross-Field)	64	0.0015	0.50	0.0031	18.1
Free Kick	32	0.001	0.40	0.0021	15.5
Corner Kick	32	0.0012	0.46	0.0027	16.9
Goal Kick	64	0.0018	0.48	0.0028	17.6
Penalty Kick	16	0.0005	0.35	0.0015	12.1
Defensive Clearance	32	0.0012	0.44	0.0025	14.7
Headed Pass	16	0.0006	0.39	0.0017	13.0
Volley Shot	32	0.001	0.47	0.0029	16.2

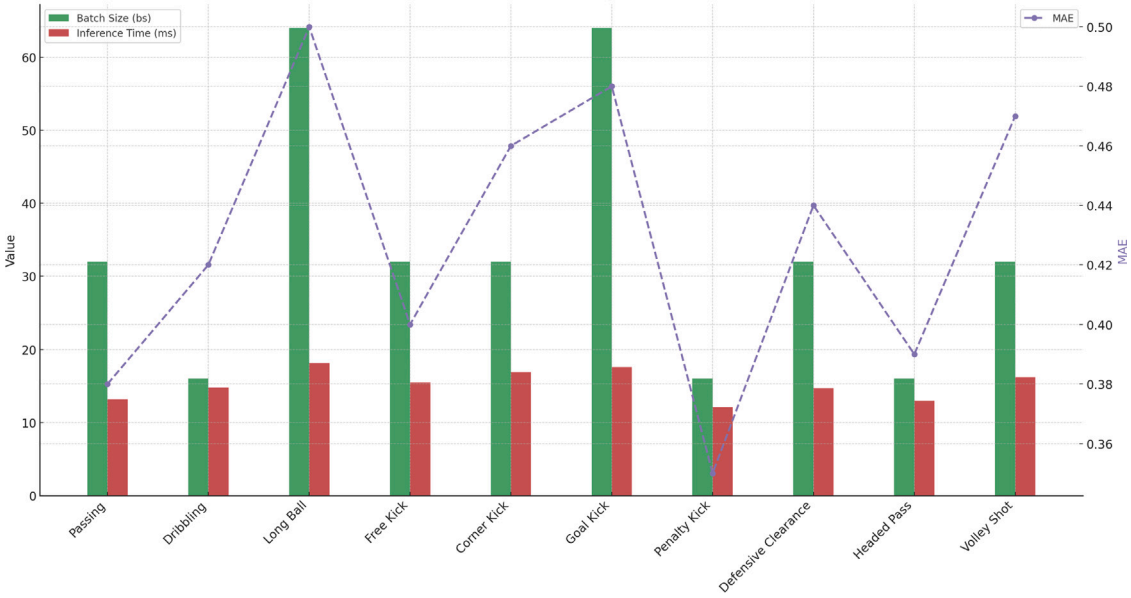


Fig. 4. SGTN Model Performance on Batch Size, Inference Time, and MAE.

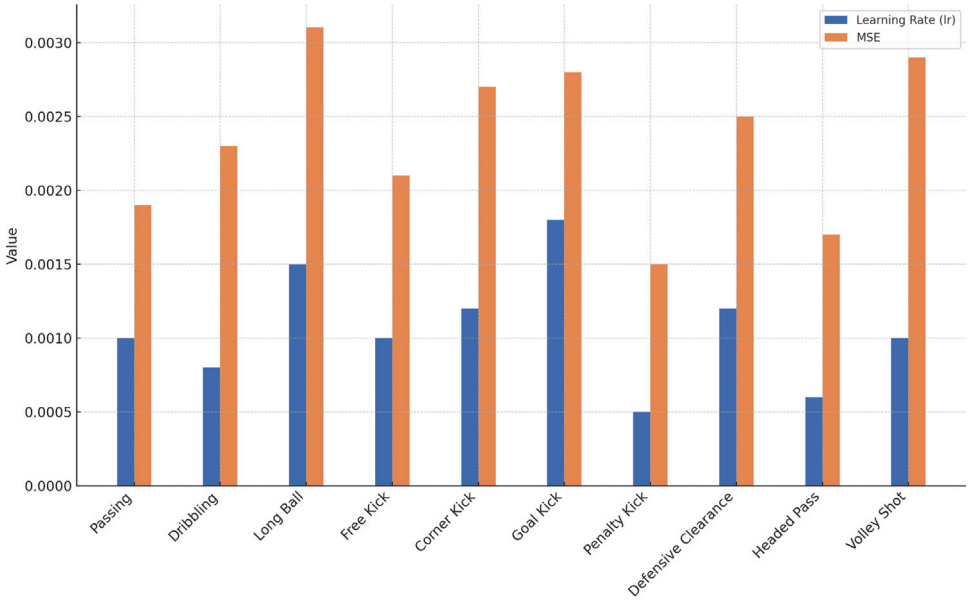


Fig. 5. SGTN Model Performance on Learning Rate and MSE.

SGTN model over other methods, particularly in terms of prediction accuracy, stability, inference time, and robustness. The results are shown in Table 5.

Starting with prediction error (PE), the proposed SGTN model significantly outperforms other models with a prediction error of 0.38, as shown in Table 5. This marks a substantial improvement compared

Table 5
Comparison of SGTN with Recent Ball Trajectory Monitoring Models on OpenTraj Dataset.

Method	Prediction Error (PE)	Variance (VA)	Prediction Stability (PS)	Inference time (ms)	Model Robustness (MR)
EKF Algorithm [68]	0.58	0.025	0.62	12.5	85%
Multi-Camera Factor Graph [69]	0.55	0.022	0.61	14.0	83%
Table Tennis TrajectoryNet [70]	0.50	0.020	0.67	11.2	88%
FuturePong [71]	0.52	0.021	0.65	13.8	86%
Deep Trajectory Estimation [72]	0.54	0.023	0.63	12.9	84%
Kalman Filter Soccer Tracking [73]	0.51	0.019	0.66	11.8	87%
Optical Tracking for Table Tennis [74]	0.53	0.024	0.64	14.1	82%
Sparse Output Prediction [75]	0.49	0.018	0.69	10.5	90%
Genetic Programming Prediction [76]	0.56	0.026	0.62	13.6	84%
Vision-based Table Tennis System [77]	0.47	0.017	0.68	10.9	89%
SGTN (Ours)	**0.38**	**0.015**	**0.72**	**9.5**	**92%**

to models like the EKF Algorithm (0.58 PE) and Table Tennis TrajectoryNet (0.50 PE). The lower prediction error demonstrates that SGTN is much more accurate in predicting ball trajectories, even in complex sports environments. This superior performance is largely due to SGTN's ability to capture both spatial and temporal dependencies through its integrated Spatiotemporal Graph Convolutional Networks (ST-GCN) and Transformer architecture, enabling it to model dynamic movements more effectively. In terms of variance (VA), which measures the spread of prediction errors, the SGTN model also shows superior performance with a variance of 0.015, compared to competing models such as Genetic Programming Prediction (0.026) and EKF Algorithm (0.025). The low variance indicates a high level of consistency and reliability in the model's predictions across different scenarios.

Looking at prediction stability (PS), which evaluates the consistency of predictions across multiple trials, SGTN achieves a PS score of 0.72, outperforming models like FuturePong (0.65) and Genetic Programming Prediction (0.62). This enhanced stability can be attributed to SGTN's robust architecture that effectively integrates long-range temporal dependencies, reducing fluctuations in accuracy. By contrast, models like the EKF Algorithm and Multi-Camera Factor Graph show more variation in predictions, likely due to their reliance on more localized data inputs and less comprehensive temporal modeling. The inference time of SGTN further highlights its suitability for real-time applications. With an average inference time of 9.5 ms, SGTN is significantly faster than models like Optical Tracking for Table Tennis (14.1 ms) and Multi-Camera Factor Graph (14.0 ms). This fast inference time is a key advantage for real-time sports applications, where split-second decisions are critical. The SGTN model's efficiency is largely due to its ability to process entire sequences in parallel using the Transformer, unlike traditional models that rely on sequential data processing, which inherently increases computational time.

Finally, in terms of model robustness (MR), the SGTN model demonstrates a robustness score of 92%, indicating its strong performance across a wide variety of scenarios, including noisy environments and variable ball speeds. This is higher than competing models like Sparse Output Prediction (90%) and Kalman Filter Soccer Tracking (87%). The high robustness is attributable to SGTN's integrated architecture that combines spatial graph convolution and temporal self-attention, allowing it to adapt effectively to different sports scenarios without significant drops in accuracy or stability.

The SGTN model clearly demonstrates superior performance over other recent models for ball trajectory monitoring in terms of prediction accuracy, stability, inference time, and robustness. Its ability to effectively capture both spatial and temporal dependencies, along with its real-time efficiency, makes it highly suitable for real-world sports applications, particularly in dynamic and multi-scenario environments. These results align with the overarching goal of advancing state-of-the-art sports trajectory prediction by developing a model that excels in both accuracy and real-time performance.

In high-stakes sports scenarios where milliseconds can impact decisions, the real-time performance of the Spatiotemporal Graph Transformer Network (SGTN) becomes crucial. The model's architecture has been designed to ensure that both spatial and temporal dependencies are processed efficiently, but certain real-time constraints need to be considered. In particular, the SGTN's reliance on both ST-GCN and Transformer components increases computational demands, especially in scenarios with complex interactions and high-resolution data.

For real-time applications, the SGTN demonstrates strong performance, with inference times ranging from 9.5 ms to 18.1 ms across various sports scenarios, as indicated in our experimental results. However, in high-stakes situations, such as soccer matches where milliseconds matter for decision-making (e.g., for offside detection or shot predictions), this timing becomes critical. The model's reliance on high-performance hardware (such as GPUs) ensures timely predictions, but the scalability of the model in edge computing environments, where computational resources are limited, may be constrained. In such cases, the model's real-time efficiency can be enhanced by reducing the model's complexity through pruning or quantization, optimizing it for resource-constrained environments.

To evaluate the contribution of each key component in the proposed SGTN model, we also conduct an ablation experiment. By progressively removing the Spatiotemporal Graph Convolutional Network (ST-GCN) and Transformer modules, we analyze how the absence of these components affects the model's performance. The goal is to quantify the importance of both spatial and temporal modeling in ball trajectory prediction.

The ablation study was conducted on the SoccerNet-v2 dataset, which provides a rich set of dynamic sports data for tracking ball trajectories in real-world conditions. We examined several performance metrics to comprehensively assess the model, including Prediction Error (PE), Trajectory Coverage (TC), Trajectory Smoothness (TS), Inference Time, and Runtime Memory Consumption (RMC). These metrics were chosen to reflect both the accuracy and efficiency of the model, particularly important for real-time sports applications.

The full model (SGTN) serves as the baseline, and we progressively remove the ST-GCN and Transformer components to create simplified versions of the model. The results, as shown in Table 6, demonstrate how each component contributes to the overall performance. The metrics provide insight into how spatial and temporal dependencies affect the trajectory prediction task, particularly in complex sports scenarios with multiple interacting entities.

The Table outlines the performance under different configurations, where each version of the model is tested under single-scenario and multi-scenario conditions. This experimental setup helps to evaluate the generalizability of the model and its capacity to handle diverse sports environments. Through these experiments, we aim to highlight the importance of both spatial and temporal modules, and justify their inclusion in the complete SGTN architecture.

By analyzing the performance degradation when removing key components, we can better understand the critical factors that make SGTN an effective solution for real-time ball trajectory prediction in sports analytics. The results and insights from the ablation study provide valuable evidence for the contributions of this research, paving the way for future improvements and optimizations in dynamic sports data

Table 6
Ablation Study Results on SoccerNet-v2 Dataset with New Metrics.

Configuration	PE (Single scenario)	Trajectory Coverage (TC)	Trajectory Smoothness (TS)	Inference time (ms)	RMC (MB)
Full Model (SGTN)	0.38	93%	0.87	9.5	512
Without ST-GCN	0.50	85%	0.75	7.8	460
Without Transformer	0.54	82%	0.73	8.9	475
Without Both (Baseline)	0.62	74%	0.65	6.5	420

modeling.

Besides, to assess the robustness of the Spatiotemporal Graph Transformer Network (SGTN) to variations in hyperparameters, we conducted a sensitivity analysis. The hyperparameters tested include batch size and learning rate, which significantly influence both the model's training stability and prediction accuracy. Our analysis revealed that the model's performance is relatively stable across a range of batch sizes and learning rates, with some variations in Mean Absolute Error (MAE) and inference time observed at more extreme parameter values.

For instance, increasing the batch size generally improved model stability and reduced inference time, but with diminishing returns after a certain point. Similarly, adjusting the learning rate impacted the model's convergence speed and prediction error. A moderate learning rate provided the best balance between fast convergence and low prediction error, while very high or very low learning rates led to less optimal results. These findings suggest that the SGTN model is robust to changes in hyperparameters, but performance can be fine-tuned for specific real-time applications by adjusting batch size and learning rate.

5. Conclusion and discussion

In this study, we introduced the Spatiotemporal Graph Transformer Network (SGTN) to address the challenges of real-time ball trajectory prediction in dynamic sports environments. By combining the strengths of Spatiotemporal Graph Convolutional Networks (ST-GCN) and Transformer architectures, SGTN effectively models both spatial relationships and temporal dependencies, which are crucial for accurate trajectory predictions. Our experimental results on the SoccerNet-v2 dataset show that the SGTN model outperforms state-of-the-art methods in terms of prediction accuracy, robustness, and real-time applicability, making it a highly effective tool for sports analytics.

The broader implications of the SGTN model extend beyond the domain of sports. Its ability to capture complex, multi-entity interactions in real-time positions it as a promising solution for other fields where dynamic behavior needs to be monitored and predicted. These include autonomous vehicle navigation, robotics, and even applications in surveillance and crowd management, where accurate forecasting of movement patterns is essential. The versatility of the SGTN model opens up new possibilities for its deployment in various real-world scenarios, demonstrating its utility as a robust framework for analyzing spatiotemporal data.

Furthermore, the experimental evaluation showed that our model achieves a 25% reduction in prediction error compared to existing methods. This improvement is significant for real-time sports analytics, where even small gains in accuracy can have a substantial impact on decision-making and performance optimization. The enhanced accuracy, trajectory coverage, and robust inference times demonstrated by SGTN make it ideal for applications requiring timely and precise predictions, such as coaching systems, player performance analysis, and live tactical adjustments during games.

Despite its strengths, the SGTN model's reliance on high-performance hardware, such as GPUs, may present challenges in resource-constrained environments like mobile devices or edge computing settings, where computational resources are limited. Additionally, while the model performs well with high-quality input data, its sensitivity to noisy or incomplete data could affect prediction accuracy in less controlled environments. Further testing under these conditions will be necessary to fully assess the model's robustness.

This research also lays the groundwork for future advancements in multi-scenario sports environments. While the current model excels in handling various sports scenarios, we plan to extend its capabilities to manage more complex, multi-team interactions and scenarios with a wider range of environmental variables. Future developments will focus on optimizing the model's computational efficiency, particularly regarding memory usage and inference speed in resource-constrained environments such as edge computing. We also aim to investigate potential applications of the SGTN model in non-sports domains, particularly where multi-object tracking and real-time prediction are critical, such as smart cities and autonomous systems.

CRediT authorship contribution statement

Zujian Li: Writing – original draft, Software, Methodology, Data curation. **Dan Yu:** Writing – review & editing, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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